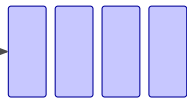


Volumetric attention branch
(CBAM-3D ResNet-34)



$$f_{\text{cbam}} \in \mathbb{R}^{512}$$

Per-IVD
volume

$9 \times 112 \times 224$



MRI slices

**Multimodal
branch**
(BiomedCLIP
ViT-B/16)

slice
attention
pool

$$f_{\text{bmc}} \in \mathbb{R}^{512}$$

concat
1024



Fusion head

(MLP $1024 \rightarrow 768 \rightarrow 512$, L2)

$$\hat{f}_{\text{img}}$$

cosine
similarity

text prompts $\{q_k\}$
 $\rightarrow$ multimodal text encoder

Trained label set
(e.g. RSNA severity)

New label set,
zero-shot
(e.g. SPIDER)